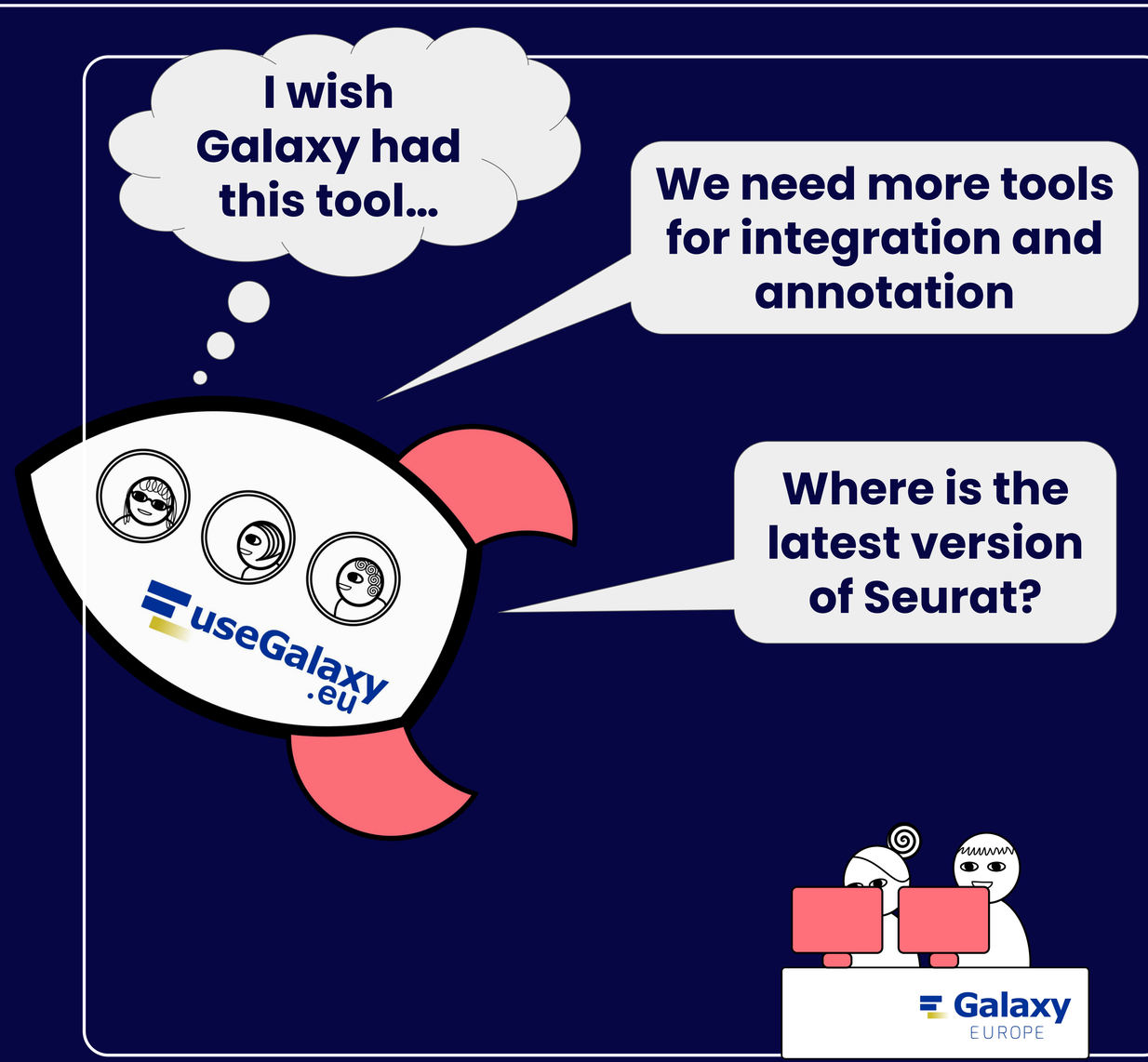
Giving the Single Cell Community the Tools It Needs on Galaxy

An ELIXIR Staff Exchange Project



The Project

Galaxy provides a selection of single cell tools, including commonly used pipelines such as **Seurat** and **Scanpy**. However, there are still plenty of single cell tools that aren't available on the main servers – with new tools being created all the time. One barrier to making more of these tools available is that while Galaxy users (and potential users) might know which tools they want to use, they may not know how to get them onto the platform. Meanwhile, the Galaxy developers and maintainers who know how to wrap tools aren't necessarily aware of which of the thousands of available tools users actually need, even if they have enough time to work on them. A six-month long project was initiated as part of the **ELIXIR Staff Exchange Programme** to bring these two groups together and expand the single cell tools available on Galaxy. The ongoing project enabled a PhD student from a group of Galaxy users at **The Open University** in the UK to join the **Galaxy EU** team in Freiburg to learn how to wrap tools and become a user-developer.

USERS

know which tools and features they need

- Commonly used pipelines like Scanpy and Seurat are essential - they need to be reliable, up-to-date and easy to use
- Data integration is becoming more important as single cell datasets increase in size and complexity
- Many new tools focus on cell annotation now that large reference datasets are available

& can learn to wrap tools for themselves

KNOWLEDGE →

← **SKILLS**

DEVELOPERS

have skills to wrap the tools that users need

- Setting up Planemo, Github accounts etc - this can be a major barrier for new developers
- Signposting tutorials for contributors (GTN, Planemo) and tool documentation
- Reviewing wrappers and helping fix errors to ensure new developers don't get stuck
- Sharing wrappers through the Tool Shed or repos like tools-iuc to ensure they will be used

& can provide guidance on wrapping

Galaxy Contribution

Updating the Seurat Wrapper

The latest version of Seurat was a common request from users

- New wrappers are being created for **Seurat version 5**
- Tools are being reorganised across several wrappers corresponding to different stages of analysis to make them **easier to find and use**
- More requested features, such as **integration** tools, will also be added to the pipeline

How Should We Make Use of Users?

Listen to Users

Provide multiple channels to capture a broad range of views

Find them here:

Single Cell Galaxy

- **Tool Requests Form**
- **Galaxy help forums**
- **Galaxy Matrix chat**
- **Single Cell Groups**
  - SPOC (Single cell & sPatial Omics CoP)
  - ELIXIR Single Cell Omics Community
- **Github issues**
- **Direct consultation & collaboration**

ELIXIR

Train More User - Developers

Support users with placements & mentorships to benefit everyone

Users need **time** to focus on **skill-building** & help from devs to become **user-devs**

- More developers means problems get fixed faster
- User-devs can draw on experience to shape features & improve usability
- Users know the latest trends & now can add the new tools they need

Devs gain time to focus on more complicated issues

Marisa Loach
The Open University
marisa.loach@open.ac.uk

Freiburg Galaxy Team
contact@usegalaxy.eu
https://bawü.social/@galaxyfreiburg

Galaxy EUROPE · elixir · The Open University